# Lab 0 ticket

**Computers Speak Vectors — Friday, August 28**

**Lab 0 — Computers Speak Vectors** · Fri Aug 28 · Laptop required · Handwrite on this hard copy · Get initials after check-in

**Name:** ______________________________ **Language (circle):** Octave / Python

## New syntax (fill as you go)

| I needed to... | Command / pattern (your language) |
|---|---|
| Create vector `x` | |
| Length `n` | |
| Print 3rd Boyd entry | |
| Change an entry | |
| Scale / add vectors | |
| Plot | |

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**Data:** $x = (52, 51, 53, 55, 58, 60)$ °F, hours 7 a.m.–noon. Boyd $x_1 \ldots x_6$. Octave: 1-based. Python: 0-based. Do **C0.1–C0.6 in order.**

**C0.1 — Enter and index.** Create `x`. Print $n$, first, last, Boyd $x_3$.

**Expected:** $n = 6$, first 52, last 60, third 53.

**Yours:** $n =$______ first=______ last=______ third=______

Command that prints the third entry: ______________________________

$x_3$ means (quantity + units): ______________________________

**C0.2 — Modify.** Set 9 a.m. to 54, noon to 61. Print new `x`.

**Expected:** $(52, 51, 54, 55, 58, 61)$. **Yours:** ______________________________

**C0.3 — Scale.** `doubled = 2*x` (use corrected `x`). Print `doubled`.

**Expected:** $(104, 102, 108, 110, 116, 122)$.

Assignment line: ______________________________

Meaning of $\times 2$ here: ______________________________

**C0.4 — Add a vector.** `ones6` = six ones; `warmer = x + 5*ones6`. Print `warmer`.

**Expected:** $(57, 56, 59, 60, 63, 66)$. **Yours:** ______________________________

Why not use math-course "`x+5`" as vector addition? ______________________________

**C0.5 — Plot** corrected `x`, `2*x`, and `-x` vs $i = 1..6$ (legend).

What did $\times(-1)$ do in the picture? ______________________________

**C0.6 — Predict, then check.** Before running: Octave `v=[10,20,30]; v(2)` or Python `v=np.array([10,20,30]); print(v[1])`.

**Expected:** 20. Prediction: ______ Actual: ______ Boyd index: $v$____

**Stuck on (optional):** C0.___ ______________________________

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